

Logos Experiment 9 Report

Wider-Shallower Transformer with Weight Tying and SwiGLU Feed-Forward Network

1. Experiment Purpose

Experiment 9 combines the strongest efficiency ideas observed so far in the Logos project. The goal was to test whether a wider-shallower architecture could preserve the speed benefits of Experiment 8 while adding the parameter and storage savings of weight tying and the quality improvement of SwiGLU feed-forward layers.

This experiment is important because the central Logos objective is not only to lower validation loss, but to improve capability per parameter, checkpoint byte, training second, latency unit, and watt.

2. Hypothesis

The hypothesis was that combining a wider-shallower Transformer, weight tying, and SwiGLU would create a stronger efficiency-focused language model than the original baseline. Specifically, it was expected to reduce parameters and checkpoint size, improve training speed, and maintain or improve validation loss compared to the baseline.

3. Model Configuration

Component	Value
Run name	logos_exp09_wider_shallower_weight_tying_swiglu
Architecture	Decoder-only causal Transformer
Architecture shape	Wider-shallower
Layers	4
Embedding dimension	288
Attention heads	9
Head dimension	32
Context length	256 tokens
Batch size	64
Tokens per step	16,384
FFN type	SwiGLU
SwiGLU hidden size	768
Normalization	LayerNorm
Weight tying	Enabled
Tokenizer	GPT-2 tokenizer
Dataset	OpenWebText subset
Training iterations	1500

4. Method Change

This experiment combines three changes:

Method	Purpose
Wider-shallower architecture	Reduce sequential depth and improve training/inference speed
Weight tying	Share input embedding and output LM head weights to reduce parameters and checkpoint size
SwiGLU FFN	Use a modern gated feed-forward layer to improve representational quality

5. Experiment 9 Results

Metric	Value
Parameters	18,542,688
Best iteration	1500
Best validation loss	5.7388
Final validation loss	5.7490
Final validation perplexity	313.87
Checkpoint size	70.76 MB
Peak VRAM	10.54 GB
Training time	132.14 sec (~2.20 min)
Training speed	~187,352 tokens/sec
Generation speed	78.69 tokens/sec
Generation latency	12.71 ms/token
Average GPU power	250.83 W
Estimated GPU energy	9.06 Wh

6. Comparison with Key Runs

Metric	Baseline	Exp 5: WT + SwiGLU	Exp 8: Wider-Shallower	Exp 9
Parameters	32.16M	19.24M	33.07M	18.54M
Best val loss	5.7716	5.7230	5.7491	5.7388
Final val loss	5.7634	5.7247	5.7620	5.7490
Final perplexity	318.42	306.36	318.00	313.87
Checkpoint size	122.71 MB	73.44 MB	126.15 MB	70.76 MB
Peak VRAM	10.84 GB	11.18 GB	10.43 GB	10.54 GB
Training time	156.42 sec	157.07 sec	138.51 sec	132.14 sec
Training speed	~158k tok/s	~157k tok/s	~179k tok/s	~187k tok/s
Generation latency	10.20 ms/tok	15.04 ms/tok	8.53 ms/tok	12.71 ms/tok
Energy	10.78 Wh	10.49 Wh	9.63 Wh	9.06 Wh

7. Interpretation

Experiment 9 is the strongest efficiency-focused model so far. It achieved the lowest parameter count, smallest checkpoint size, fastest training speed, lowest energy usage, and better validation loss than the original baseline. This makes it highly valuable for the Logos goal of improving capability per resource.

However, it did not surpass Experiment 5 in raw validation quality, and it did not surpass Experiment 8 in generation latency. Therefore, Experiment 9 is not the best model for every metric, but it is the best overall efficiency model when parameter count, checkpoint size, energy, and speed are prioritized.

8. Strengths

Strength	Evidence
Lowest parameter count so far	18.54M parameters
Smallest checkpoint so far	70.76 MB
Fastest training so far	~187k tokens/sec
Lowest energy usage so far	9.06 Wh
Better than baseline quality	Final val loss improved from 5.7634 to 5.7490
Better than baseline perplexity	Perplexity improved from 318.42 to 313.87

9. Weaknesses

Weakness	Explanation
Not best validation quality	Experiment 5 still had better final val loss: 5.7247
Not best generation latency	Experiment 8 had better latency: 8.53 ms/token
Quality tradeoff vs Exp 5	The wider-shallower structure improves efficiency but slightly weakens quality compared to Exp 5

10. Scientific Conclusion

Experiment 9 successfully demonstrates that architectural shape optimization can combine with weight tying and SwiGLU to produce a more resource-efficient language model. Compared with the original baseline, it reduced parameters from 32.16M to 18.54M, reduced checkpoint size from 122.71 MB to 70.76 MB, improved training speed from about 158k tokens/sec to about 187k tokens/sec, and reduced estimated energy from 10.78 Wh to 9.06 Wh. It also slightly improved validation loss and perplexity compared to the baseline.

The clean conclusion is: Experiment 9 is the best efficiency-focused Logos model so far, while Experiment 5 remains the best quality-per-storage model.

11. Current Model Ranking

Category	Best Run	Reason
Best raw quality	Experiment 5	Lowest final validation loss and perplexity
Best efficiency overall	Experiment 9	Lowest parameters, smallest checkpoint, fastest training, lowest energy
Best generation latency	Experiment 8	Lowest measured ms/token among key runs
Best baseline comparison variant	Experiment 9	Improves quality and efficiency over baseline

12. One-Line Summary

Experiment 9 shows that a wider-shallower decoder-only Transformer with weight tying and SwiGLU can become the most efficient Logos variant so far, reducing model size and energy while maintaining better-than-baseline language modeling quality.

Prepared as part of the Logos language model efficiency experiment series.